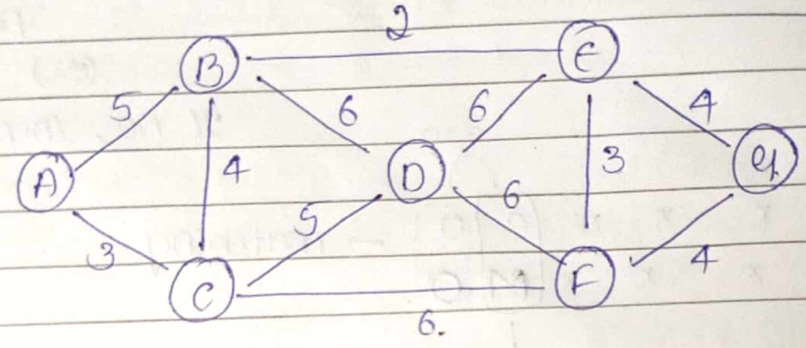
⟹ Spanning tree (Kruskal)

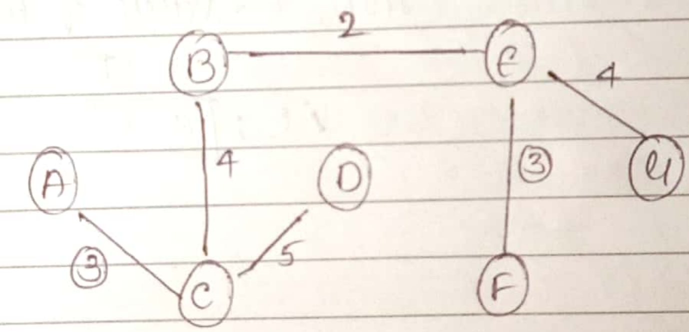
» min → max (edge cost)

» No cycle formation.

Ex:



⇓



Total edge cost = 2 + 3 + 3 + 4 + 4 + 5

= 21

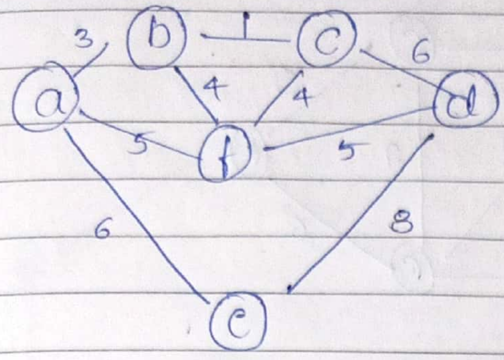
Total not edge cost = 33. (Remaining)

not edged = 6

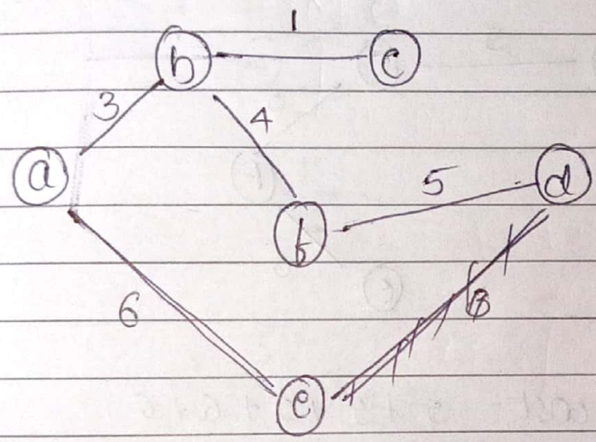
max edge cost = 6

edged = (n-1) = 6

Ex:



⇓



Total edge cost = $3 + 1 + 4 + 5 + \cancel{8}\,6$

$= \cancel{16}\ 19$

Total not edge cost = $6 + 8 + 5 + 4$

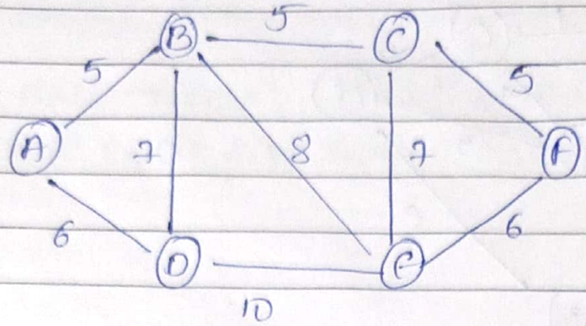
$= 23$

edged = $(n-1) = 5$.

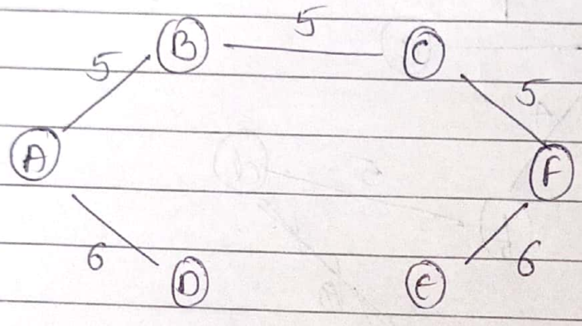
not edged = 4

max edge = 8.

→



⇓



Total edge cost = 5 + 5 + 5 + 6 + 6
= 27.

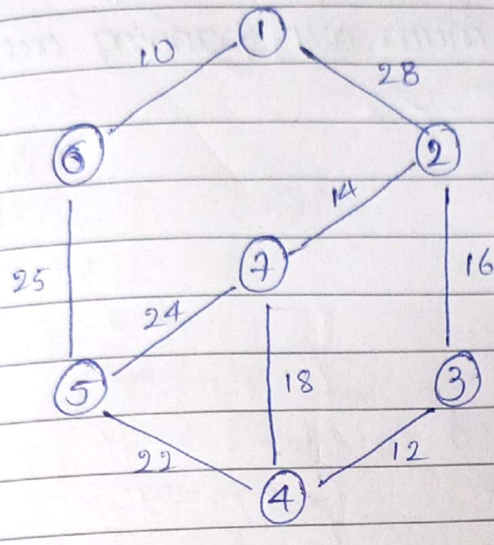
Total not edge cost = 7 + 8 + 7 + 10
= 32.

edged = (n−1) = (5)

Total not edged = 4

max edge cost = 10.

Spanning tree (Prims)

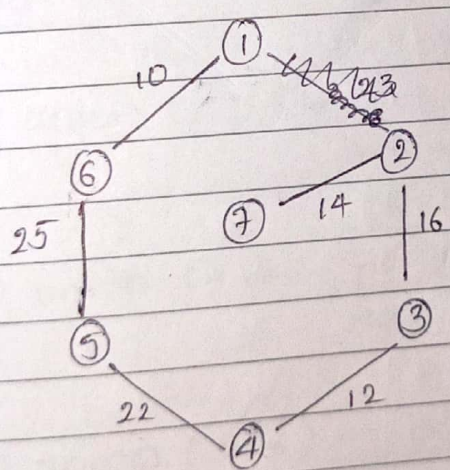


① min edge cost

② min edge cost from 1 and 6

③ repeat with all edge.. incdm connected.

⇓



Total edge cost = 10 + 25 + 22 + 12 + 16 + 14
= 99.

Total not edge cost = 28 + 24 + 18 +
= 70

edged = (n-1) = 7.

not edged = 3

max cost = 28.

⇒ Kruskal algorithm

//Input : a connected weighted graph.
//Output : a minimum cost spanning tree.

Scanned with CamScanner